Team 06

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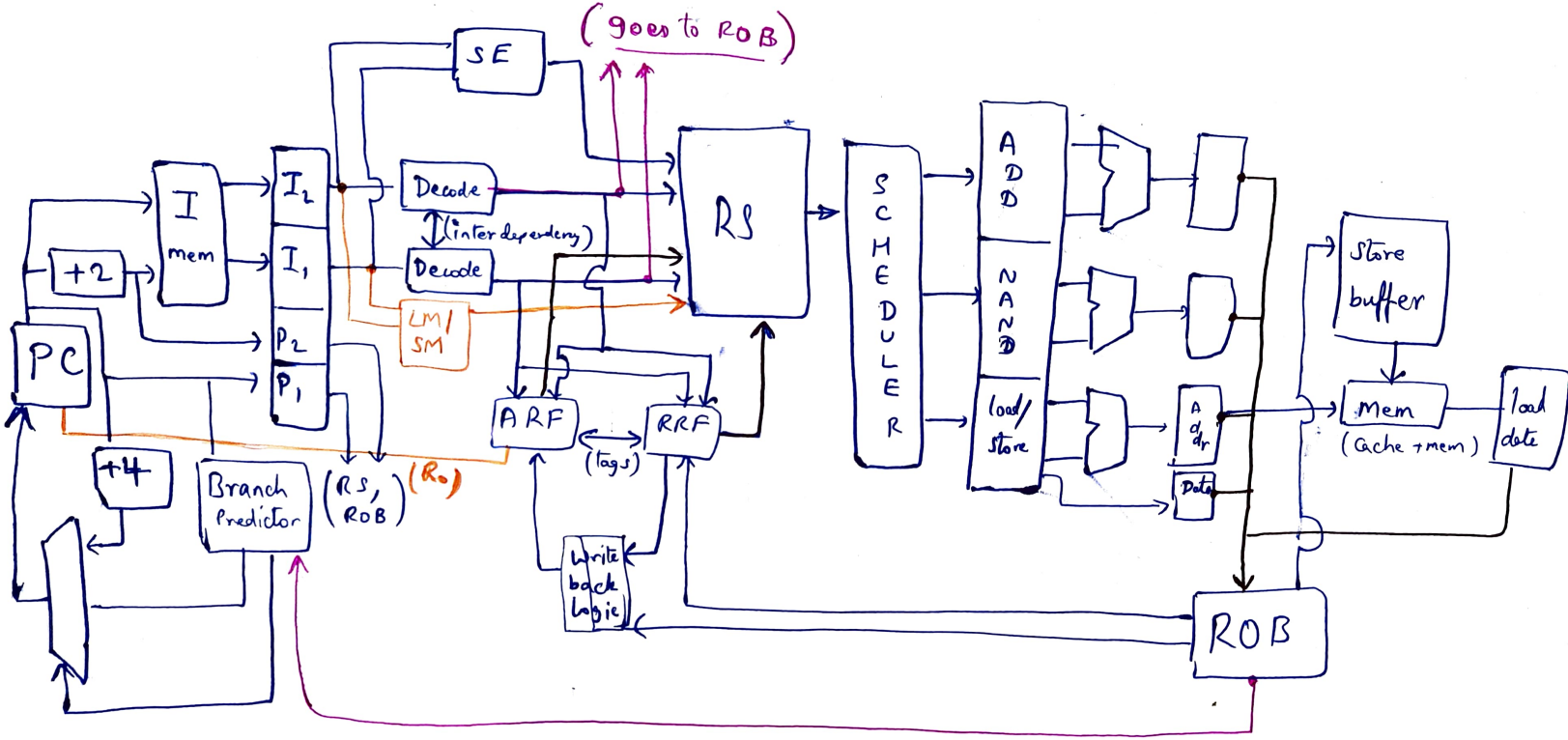
(goes to ROB)
SE
I mem
+2
PC
+4
Branch Predictor
I_2
I_1
P_2
P_1
Decode
(inter dependency)
Decode
LM/SM
(RS, ROB)
(R_0)
ARF
RRF
(tags)
Write back logic
RS
SCHEDULER
ADD
NAND
load/store
Adr
Data
Store buffer
Mem
(cache + mem)
load data
ROB

ARF (8 entries)

Busy	Value	Tag
1 bit	16 bit	5 bit

RRF (32 entries)

Busy	Value	Valid
1 bit	16 bit	1 bit

Reservation station (32 entries)

Busy	Control	OPR1	Valid 1	OPR2	Valid 2	Ready	Renamed Register (Destination)
1	4+1+2 =7	16	1	16	1	1	5

Reorder Buffer ROB (64 entries)

Busy	I-P	Rdes	Renamed Reg	Spec	Exe	Issue	Completed	Valid	S_Addr	S_Data	Branch predicted
1	16	3	5	1	1	1	1	1	16	16	

☆ Points regarding implementation:

- ROB is implemented as a FIFO
- ARF and RRF are implemented separately

  A Queue (FIFO) is used to ~~access~~ track the free registers of the RRF

- Three execution units (functional units)
  - ADD Exe - 1 cycle
  - NAND Exe - 1 cycle
  - Load / Store Exe - 2 cycle (address + mem access)

- Branch prediction:
  - Branch predicted bit in ROB is used to ~~cross~~ ~~ensure~~ check the ~~targeted and~~ predicted and taken branch are same.

- RS ⇒ 32 entries (rows)

  ROB ⇒ 64 entries.